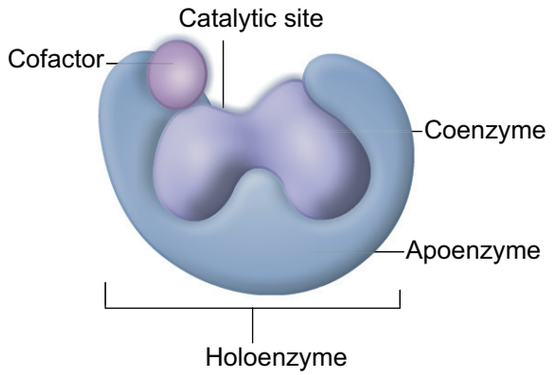



**Figure 8.19:** Enzyme components

- **Inorganic ions** help to increase the rate of reaction catalysed by enzymes. Example: Salivary amylase activity is increased in the presence of chloride ions.
- **Prosthetic groups** are organic molecules that assist in catalytic function of an enzyme. Flavin adenine dinucleotide (FAD) contains riboflavin (vit B2), the function of which is to accept hydrogen. 'Haem' is an iron-containing prosthetic group with an iron atom at its centre.
- **Coenzymes** are organic compounds which act as cofactors but do not remain attached to the enzyme. The essential chemical components of many coenzymes are vitamins. Eg. NAD, NADP, Coenzyme A, ATP




**Ribozyme - Non Protein Enzyme** - A Ribozyme, also called as catalytic RNA; is a ribonucleic acid that acts as enzyme. It is found in ribosomes

### 8.6.4 Classification of Enzymes

Enzymes are classified into six groups based on their mode of action.

| Enzymes | Mode of action | General scheme of reaction | Example |
|---|---|---|---|
| Oxidoreductase | Oxidation and reduction (redox) reactions | $A_{red} + B_{ox} \longrightarrow A_{ox} + B_{red}$ | Dehydrogenase |
| Transferase | Transfer a group of atoms from one molecule to another | $A - B + C \longrightarrow A + C - B$ | Transaminase, phosphotransfer-ase |
| Hydrolases | Hydrolysis of substrate by addition of water molecule | $A - B + H_2O \longrightarrow A - H + B - OH$ | Digestive enzymes |
| Isomerase | Control the conversion of one isomer to another by transferring a group of atoms from one molecule to another | $A - B - C \longrightarrow A - C - B$ | Isomerase |
| Lyase | Break chemical bond without addition of water | $A - B \longrightarrow A + B$ | Decarboxylase |
| Ligase | Formation of new chemical bonds using ATP as a source of energy | $A + B + ATP \longrightarrow A - B + ADP + Pi$ | DNA ligase |



XIth_BIO-BOTANY_Ch_8-EM.indd 162 5/17/2020 11:22:34 PM